Massey University
College of Sciences

161.304: Statistical Modelling

Assignment 1, 2024

GENERAL INFORMATION: This assignment is assessed. It carries weight 25% towards your final mark for the 161.304 paper. Work should be submitted electronically via the Stream site for 161.304, preferably as a PDF document with all your works, including your answer, R code, and any references used.

DEADLINE: Your work for this assignment must be submitted by **11.55pm on Friday 16 August**. Late penalty - 0 mark unless prior arrangement.

PLAGIARISM: The work that you submit must be your sole effort (i.e. not copied from anyone else). If you are found guilty of plagiarism you will be penalized. Happy for you to use ChatGPT to improve your code or your English, only state how you have used it if you have. It is an amazing tool and much can be learned form its use. Automatically generated solutions are still easy to spot and often weird, so please do not try to use this tool without thinking.

EXERCISE 1: How much water on Earth?

Consider again the glob tossing model you encountered in Week 2's chapter about simulations. Compute and plot the grid approximation of the posterior probability distribution for p, the proportion of water on earth.

1. For the following sets of observations: (a) W, W, W and (b) L, W, W, L, W, W, W assuming a uniform prior on p

   [6 marks]

2. Now start over assuming a prior for p that is equal to zero when $p < 0.5$ and is positive when $p \geq 0.5$. Again compute and plot the grid approximation of the posterior distribution of p for each set of observations in the previous point.

   [6 marks]

3. This is a more open-minded problem. You want to estimate the proportion of water on earth very precisely. Specifically, you want to obtain a 99% percentile (aka credible) interval for the posterior distribution of p and you want this interval to be no more than 0.05 wide; this is the distance between the lower and upper bounds of the interval. How many times will you have to toss the globe to do this? I do not necessarily require a precise quantitative answer. I'm mainly interested in your reasoning to approach this question.

   [6 marks]

EXERCISE 2: Earth versus Mars

Do exercise 2M3 of Chapter 2 of the *Statistical Rethinking* book.

[8 marks]

EXERCISE 3: Panda versus Panda

Do exercise 2H1 of Chapter 2 of the *Statistical Rethinking* book.

[10 marks]

EXERCISE 4: !Kung census predictions and modelling

1. The weights listed below were recorded in the !Kung census, but heights were not recorded for these individuals. Provide predicted heights and 89compatibility intervals for each of these individuals. That is, fill in the table below, using model-based predictions (see Chapter 4 of the *Statistical Rethinking* book).

Individual	Weight	Expected height	89% interval
1	45		
2	40		
3	65		
4	31		

   You may notice that the table is concerned with adults only.

   [8 marks]

2. Model the relationship between height (cm) and the natural logarithm of weight (log-kg): log(weight). Use the entire `Howell1` data frame, all 544 rows, adults and non-adults. Use any model type from Chapter 4 of the *Statistical Rethinking* book that you think useful: an ordinary linear regression, a polynomial or a spline. Plot the posterior predictions against the raw data.

   [10 marks]

3. Plot the prior predictive distribution for the polynomial regression model in Chapter 4. You can modify the the code that plots the linear regression prior predictive distribution. 20 or 30 parabolas from the prior should suffice to show where the prior probability resides. Can you modify the prior distributions of α, β_1, and β_2 so that the prior predictions stay within the biologically reasonable outcome space? That is to say: Do not try to fit the data by hand. But do try to keep the curves consistent with what you know about height and weight, before seeing these exact data.

   [10 marks]

EXERCISE 5: The Gamblers

Two gamblers, Algernon and Bethany, play a sequence of card games. Both of them start with \$3, and after each game the loser pays the winner \$1. The sequence of games comes to a stop when one or other play runs out of money, or when they have played 25 games (whichever comes first). Bethany is the more skilful card player and wins each game with probability 0.55. Use stochastic simulation (see Tutorial 1) to estimate the probability that (i) Algernon wins all of the Bethany's money; and (ii) that Bethany wins all of Algernon's money.

You should submit the following:

- Your estimates for (i) and (ii);
- Appropriate standard errors for your estimates;
- A copy of the R code that you use.

[11 marks]